ECE 102 Homework 3

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1. **Linear Systems**

   Determine whether each of the following systems is linear:

   (a) $y(t) = \cos(t)x(t)$

   Solution:

$$
\begin{aligned}
S(ax(t) + b\tilde{x}(t)) &= \cos(t)(ax(t) + b\tilde{x}(t)) && (1)\\
&= a\cos(t)x(t) + b\cos(t)\tilde{x}(t) && (2)\\
&= aS(x(t)) + bS(\tilde{x}(t)) && (3)
\end{aligned}
$$

   Thus, the system is linear

   (b) $y(t) = \frac{d}{dt}(\frac{1}{2}x(t)^2)$

   Solution:

$$
\begin{aligned}
S(ax(t) + b\tilde{x}(t)) &= \frac{d}{dt}\left(\frac{1}{2}(ax(t) + b\tilde{x}(t))^2\right) && (4)\\
&= \frac{d}{dt}\left(\frac{1}{2}((ax(t))^2 + (b\tilde{x}(t))^2 + 2abx(t)\tilde{x}(t))\right) && (5)\\
&= \frac{d}{dt}\left(\frac{1}{2}(ax(t))^2\right) + \frac{d}{dt}\left(\frac{1}{2}(b\tilde{x}(t))^2\right) + \frac{d}{dt}\left(2abx(t)\tilde{x}(t)\right) && (6)\\
&\neq aS(x(t)) + bS(\tilde{x}(t)) && (7)
\end{aligned}
$$

   The system is not linear

   (c) $y(t) = e^{x(t)}$

   Solution:

$$
\begin{aligned}
S(ax(t) + b\tilde{x}(t)) &= e^{ax(t)+b\tilde{x}(t)} && (8)\\
&= e^{ax(t)} \cdot e^{b\tilde{x}(t)} && (9)\\
&\neq ae^{x(t)} + be^{\tilde{x}(t)} && (10)
\end{aligned}
$$

The system is not linear

(d) $y(t) = x(t) + 2u(t+1)$

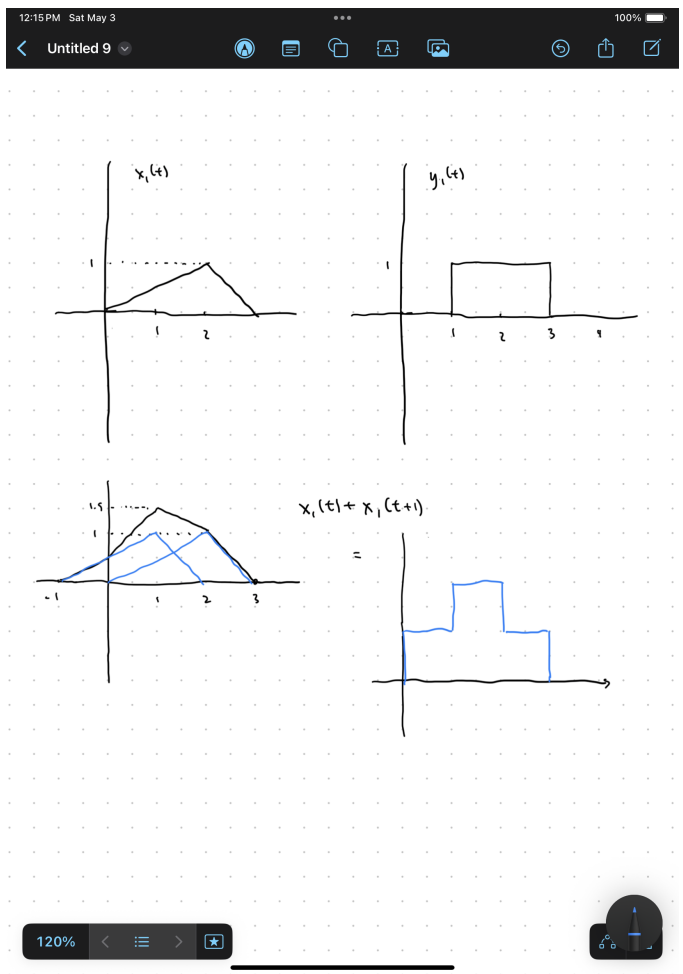
Solution:

$$S(ax(t) + b\tilde{x}(t)) = ax(t) + b\tilde{x}(t) + 2u(t+1) \tag{11}$$

The system is not linear. Additionally, $S(0) \implies$ the output should be 0 for a linear system. However, $y(t)$ contains a function $2u(t+1)$ independent of $x(t)$. Therefore the system would violate homogeneity

2. **LTI Systems**

(a) Consider an LTI (linear time-invariant) system whose response to $x_1(t)$ is $y_1(t)$, where $x_1(t)$ and $y_1(t)$ are illustrated as follows:



(b) Assume we have a linear system with the following input-output pairs:

- the output is $y_1(t) = \cos(t)u(t)$ when the input is $x_1(t) = u(t)$
- the output is $y_2(t) = \cos(t)(u(t+1) - u(t))$ when the input is $x_2(t) = \text{rect}(t + \frac{1}{2})$

Is the system time-invariant?

Solution:

Notice that $u(t+1) - u(t) = \text{rect}(t + \frac{1}{2}$. Assume $y(t) = \cos(t)x(t)$

$$H(ax(t) + b\tilde{x}(t) = \cos(t)(ax(t) + b\tilde{x}(t)) \tag{12}$$
$$= \cos(t)ax(t) + \cos(t)b\tilde{x}(t) \tag{13}$$
$$= aH(x(t)) + bH(\tilde{x}(t)) \tag{14}$$

We are given that $S(u(t)) = \cos(t)u(t)$ and $S(\text{rect}(t + \frac{1}{2}) = \cos(t)(u(t+1) - u(t))$.

Shifting the input,

$$y(t - t_0) = \cos(t)x(t - t_0) \tag{15}$$

The system is time-invariant.

3. **Convolution:**

(a) For each pair of the signals given below, compute their convolution using the flip-and-drag technique

i. $f(t) = \delta(t+1) + 2\delta(t-2), \quad g(t) = e^{-t}u(t)$

Solution:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau \tag{16}$$
$$= \int_{-\infty}^{\infty} (\delta(t+1) + 2\delta(t-2))e^{-(t-\tau)}u(t-\tau)d\tau \tag{17}$$
$$= \int_{-\infty}^{\infty} \delta(t+1)e^{-(t-\tau)}u(t-\tau)d\tau + \int_{-\infty}^{\infty} 2\delta(t-2)e^{-(t-\tau)}u(t-\tau)d\tau \tag{18}$$
$$= e^{-(t+1)}u(t+1) + 2e^{-(t-2)}u(t-2) \tag{19}$$

ii. $f(t) = 2\text{rect}(t - \frac{3}{2}), \quad g(t) = 2r(t-1)\text{rect}(t - \frac{3}{2})$

Solution:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau \tag{20}$$

$f(\tau) \neq 0 \iff \tau \in [1,2]$ and $g(t-\tau) \neq 0 \iff t - \tau \in [1,2] \iff \tau \in [t-2, t-1]$.

Thus $[1,2] \cap [t-2, t-1] \neq \emptyset \iff \max(1, t-2) < \min(2, t-1) \iff t \in (2,3)$

For $t \in (2,3)$:

$f(\tau) \neq 0 \implies f(\tau) = 2$

$g(t-\tau) \neq 0 \implies g(t) = 2(t-\tau-1)$ Then

$$(f*g)(t) = \int_{\max(1,t-2)}^{\min(2,t-1)} f(\tau)g(\tau)d\tau \quad \text{for} \quad t \in (2,3) \tag{21}$$

$$= 4\int_{\max(1,t-2)}^{\min(2,t-1)} (t-\tau-1)d\tau \tag{22}$$

Notice that we can just integrate from $t-2$ to $t-1$, because non-overlapping regions get zeroed out. Thus

$$(f*g)(t) = 4\int_{t-2}^{t-1} (t-1-\tau)d\tau \tag{23}$$

$$= 4\left[(t-1)\tau - \frac{1}{2}\tau^2\right]_{t-2}^{t-1} \tag{24}$$

$$= 4\left[(t-1)^2 - \frac{1}{2}(t-1)^2 - (t-1)(t-2) + \frac{1}{2}(t-2)^2\right] \tag{25}$$

$$= 4\left[\frac{1}{2}(t-1)^2 - (t-1)(t-2) + \frac{1}{2}(t-2)^2\right] \tag{26}$$

Thus the convolution is

$$(f*g)(t) = \begin{cases} 2\left[(t-1)^2 - 2(t-1)(t-2) + (t-2)^2\right] & t \in (2,3) \\ 0 & \text{otherwise} \end{cases} \tag{27}$$

(b) For each of the following, find a function $h(t)$ such that $y(t) = x(t) * h(t)$

i. $y(t) = \int_{t-T}^{t} x(\tau)d\tau$

Solution:

We want

$$\int_{t-T}^{t} x(\tau)d\tau = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \tag{28}$$

Therefore $h(t-\tau) = 1$ when $\tau \in [t-T, t]$ and 0 otherwise.

We have that $\tau \in [t-T, t]$

$$\iff t-T \leq \tau \leq t \tag{29}$$

$$-T \leq \tau - t \leq 0 \tag{30}$$

$$0 \leq t-\tau \leq T \tag{31}$$

Let $s = t - \tau$. Thus, we want $h(s) = 1$ for $s \in [0, T]$ and 0 otherwise.

$$h(s) = \begin{cases} 1, & s \in [0, T] \\ 0, & \text{otherwise} \end{cases} \tag{32}$$

Thus $h(t) = u(t) - u(t - T)$.

ii. $y(t) = \sum_{n=-\infty}^{\infty} x(t - nT_s)$

Solution:

We want

$$\sum_{n=-\infty}^{\infty} x(t - nT_s) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau \tag{33}$$

Notice we are adding discrete values of some periodic function $x(t)$. Therefore we want $h(t)$ to zero out all the values of $x(t)$ that aren't $t - nT_s$ for $n \in \mathbb{Z}$.

We want to "pick out" $x(t - nT_s)$ for each $n \in \mathbb{Z}$. Thus we should use the Dirac Delta function and the sifting property. Recall that

$$\int_{-\infty}^{\infty} f(\tau)\delta(\tau - a)d\tau = f(a) \tag{34}$$

Therefore,

$$\int_{-\infty}^{\infty} x(\tau)\delta(\tau - (t - nT_s))d\tau = x(t - nT_s) \tag{35}$$

Thus,

$$\int_{-\infty}^{\infty} \left(x(\tau) \sum_{n=-\infty}^{\infty} \delta(\tau - (t - nT_s)) \right) d\tau = \sum_{n=-\infty}^{\infty} x(t - nT_s) \tag{36}$$

$$\int_{-\infty}^{\infty} \left(x(\tau) \sum_{n=-\infty}^{\infty} \delta(nT_s - (t - \tau) \right) d\tau = \sum_{n=-\infty}^{\infty} x(t - nT_s) \tag{37}$$

Hence,

$$h(t - \tau) = \sum_{n=-\infty}^{\infty} \delta(nT_s - (t - \tau)) \tag{38}$$

$$\implies h(t) = \sum_{n=-\infty}^{\infty} \delta(nT_s - t) \tag{39}$$

(c) Simplify the following expressions:

i. $[\delta(t - 3) + \delta(t + 2)] * [e^{3t}u(-t) + \delta(t + 2) + 2]$

Solution:

Recall that

$$(f+k)*(g+h) = ((f+k)*g) + ((f+k)*h) \tag{40}$$
$$= (f*g) + (k*g) + (f*h) + (k*h) \tag{41}$$

Let $f = \delta(t-3), \quad k = \delta(t+2), \quad g = e^{3t}u(-t), \quad h = \delta(t+2)+2$
Then,

$$(f*g) = e^{3t-9}u(3-t) \tag{42}$$

$$(k*g) = e^{3t+6}u(-t-2) \tag{43}$$

$$(f*h) = \delta(t-3)*[\delta(t+2)+2] \tag{44}$$
$$= \delta(t-3)*\delta(t+2) + \delta(t-3)*2 \tag{45}$$
$$= \int_{-\infty}^{\infty} \delta(t-3)(\delta(t-\tau+2)d\tau + 2\int_{-\infty}^{\infty} \delta(t-3)d\tau \tag{46}$$
$$= \delta(t-1)+2 \tag{47}$$

$$(k*h) = \delta(t+2)*[\delta(t+2)+2] \tag{48}$$
$$= \delta(t+4)+2 \tag{49}$$

Thus the final expression becomes

$$e^{3t-9}u(3-t) + e^{3t+6}u(-t-2) + \delta(t-1) + \delta(t+4) + 4 \tag{50}$$

ii. $\frac{d}{dt}[(u(t)-u(t+1))*u(t-2)]$ Hint: Show first that $u(t)*u(t) = r(t)$ where $r(t)$ is the ramp function:
Solution:

$$u(t)*u(t) = \int_{-\infty}^{\infty} u(\tau)u(t-\tau)d\tau \tag{51}$$
$$= \int_{0}^{t} u(\tau)u(t-\tau)d\tau \tag{52}$$
$$= \int_{0}^{t} 1\cdot 1 d\tau \quad \text{for } t \geq 0 \tag{53}$$
$$= t \quad \text{for } t \geq 0 \tag{54}$$

If $t < 0$, then $u(t-\tau) = 0$ unless τ is very negative, which would then make $u(\tau) = 0$. Thus $u(t) * u(t) = 0$ for $t < 0$. Thus

$$u(t) * u(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases} = r(t) \tag{55}$$

Next,

$$(u(t) - u(t+1)) * u(t-2) = u(t) * u(t-2) - u(t+1) * u(t-2) \tag{56}$$
$$= r(t-2) - r(t-1) \tag{57}$$

Differentiating, we get

$$\frac{d}{dt}[r(t-2) - r(t-1)] = u(t-2) - u(t-1) \tag{58}$$

(d) Explain whether each of the following statements is true or false.

i. If $x(t)$ and $h(t)$ are both odd functions, and $y(t) = x(t) * h(t)$, then $y(t)$ is an even function.

Solution:

We want to check if $y(-t) = y(t)$

$$y(-t) = \int_{-\infty}^{\infty} x(\tau)h(\tau - t)d\tau \tag{59}$$
$$= \int_{-\infty}^{\infty} x(\tau) \cdot -h(t-\tau)d\tau \quad h(t) \text{ odd} \tag{60}$$
$$= -\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \tag{61}$$
$$= -x(t) * h(t) \tag{62}$$
$$= -y(t) \tag{63}$$

The statement is **not true**.

ii. If $y(t) = x(t) * h(t)$, then $y(2t) = h(2t) * x(2t)$

Solution:

We are given

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \tag{64}$$

LHS:

$$y(2t) = \int_{-\infty}^{\infty} x(\tau)h(2t-\tau)d\tau \tag{65}$$

Let $\tau = 2\lambda$. Then $d\tau = 2d\lambda$. Substituting, we get

$$y(2t) = \int_{-\infty}^{\infty} x(2\lambda)h(2t - 2\lambda)2d\lambda \tag{66}$$
$$= 2\int_{-\infty}^{\infty} x(2\lambda)h(2(t - \lambda))d\lambda \tag{67}$$
$$= 2(x(2t) * h(2t)) \tag{68}$$

Thus the statement is **not true**

4. **Impulse Response and LTI Systems** Consider the following three LTI systems:

- The first system S_1 is given by its input-output relationship: $y(t) = \int_{-\infty}^{t} x(\tau - t_0)d\tau$;
- The second system S_2 is given by its impulse response $h_2(t) = u(t - 2)$;
- The third system S_3 is given by its impulse response: $h_3(t) = u(t + 3)$

(a) Compute the impulse responses $h_1(t)$ of system S_1.
Solution:

$$h_1(t) = \int_{-\infty}^{t} \delta(\tau - t_0)d\tau \tag{69}$$

Thus

$$h_1(t) = \begin{cases} 1, & t \geq t_0 \\ 0, & t < t_0 \end{cases} \tag{70}$$
$$= u(t - t_0) \tag{71}$$

(b) The three systems are interconnected. Determine the impulse response $h_{eq}(t)$ of the equivalent system.
Solution:

$$S_1(x(t)) - S_3(S_2(x(t))) \tag{72}$$

Therefore

$$h_{eq} = S_1(\delta(t)) - S_3(S_2(\delta(t))) \tag{73}$$
$$= u(t - t_0) - S_3(u(t - 2)) \tag{74}$$
$$= u(t - t_0) - u(t + 3) * u(t - 2) \tag{75}$$
$$= u(t - t_0) - (t + 1)u(t + 1) \tag{76}$$

(c) Determine the response of the overall system to the input $x(t) = \delta(t) + \delta(t-3)$.

Solution:

$$h_{eq}(x(t)) = h_{eq}(\delta(t) + \delta(t-3)) \tag{77}$$

$$= h_{eq}(\delta(t)) + h_{eq}(\delta(t-3)) \tag{78}$$

$$= u(t-t_0) - (t+1)u(t+1) + u(t-t_0-3) - (t-2)u(t-2) \tag{79}$$

```
import numpy as np
import matplotlib.pyplot as plt
```

For this homework question, we will be computing the convolution of two rect signals to see what the resulting output looks like. In the code blocks below we will be creating two rect signals, and using python numpy library functions to perform the convolution. Hint: check out the numpy.convolve() function documentation.

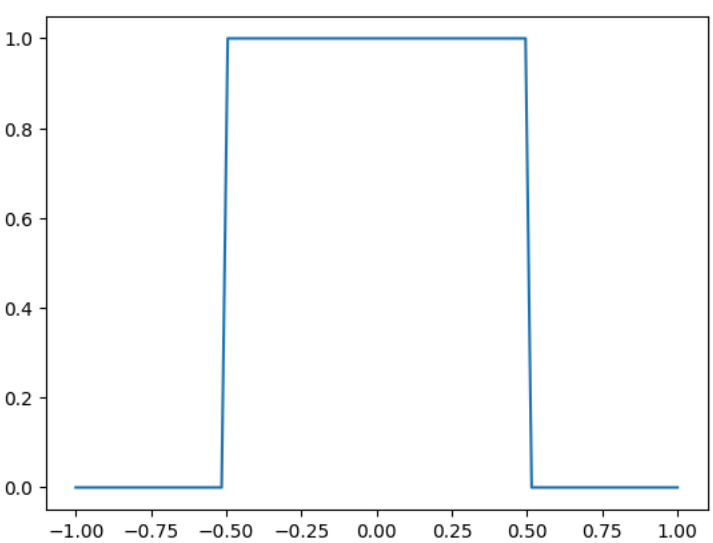
(2 points) In the cell below, we will start by making the rect function.

```
# Make the x-axis first to use when constructing the rect signal
x = np.linspace(-1, 1, 100)

# Define the width of the rectangle
width = 1

# Construct the rect function: 1 where |x| <= width/2, else 0
rect = np.where(np.abs(x) <= width / 2, 1.0, 0.0)


# Plotting code
plt.plot(x, rect)
plt.show()
```



(8 points) In the cell below, you will use what you wrote in the previous part to setup two rect signals and convolve them using numpy convolve. We will then graph it and see the signal output

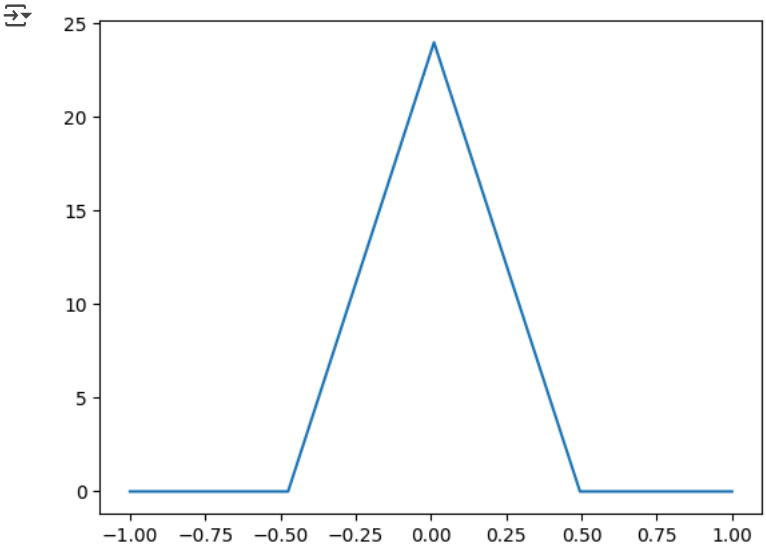
```
# Make the x-axis first to use when constructing the rect signal
x = np.linspace(-1, 1, 100)

width = 0.5

rect1 = np.where(np.abs(x) <= width / 2, 1.0, 0.0)
rect2 = np.where(np.abs(x) <= width / 2, 1.0, 0.0)

out = np.convolve(rect1, rect2, mode='same')

# Plotting code
x_plot = np.linspace(-1, 1, out.shape[0])
plt.plot(x_plot, out)
plt.show()
```



(7 points) Great! Now that you have the hang of convolutions in python, we will try something slightly more interesting. Try the convolution of 10 rect functions (rect * rect * rect * rect...). What does this output look like? Do you have any guesses as to why? (No need to respond to this last question, just an optional thing to think about if interested)

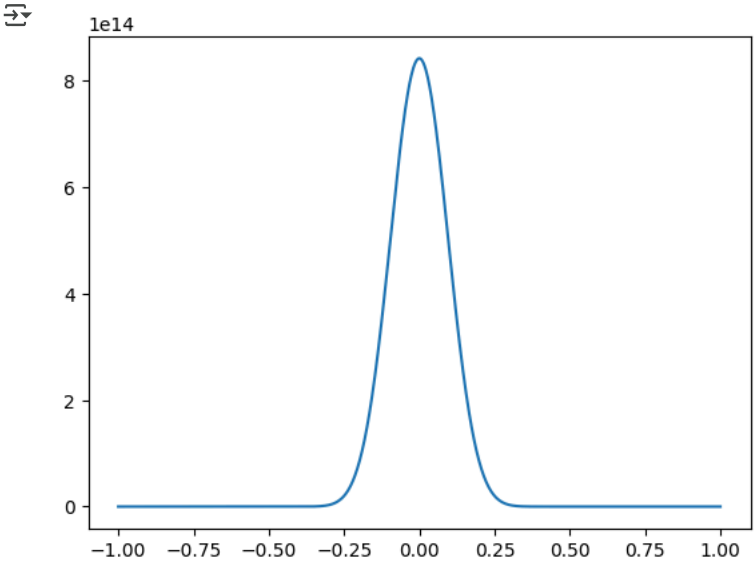
```
# Make the x-axis first to use when constructing the rect signal
x = np.linspace(-1, 1, 100)

# YOUR CODE BELOW
# Define one rect function (e.g., width 0.2)
width = 1
rect = np.where(np.abs(x) <= width / 2, 1.0, 0.0)

# Start with the initial rect
rect_out = rect.copy()

# Convolve the rect function with itself 9 more times
for _ in range(9):
    rect_out = np.convolve(rect_out, rect, mode='full')

# Plotting code
x_plot = np.linspace(-1, 1, rect_out.shape[0])
plt.plot(x_plot, rect_out)
plt.show()
```




```
#comment
```